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(54) **DOOR ASSEMBLY**

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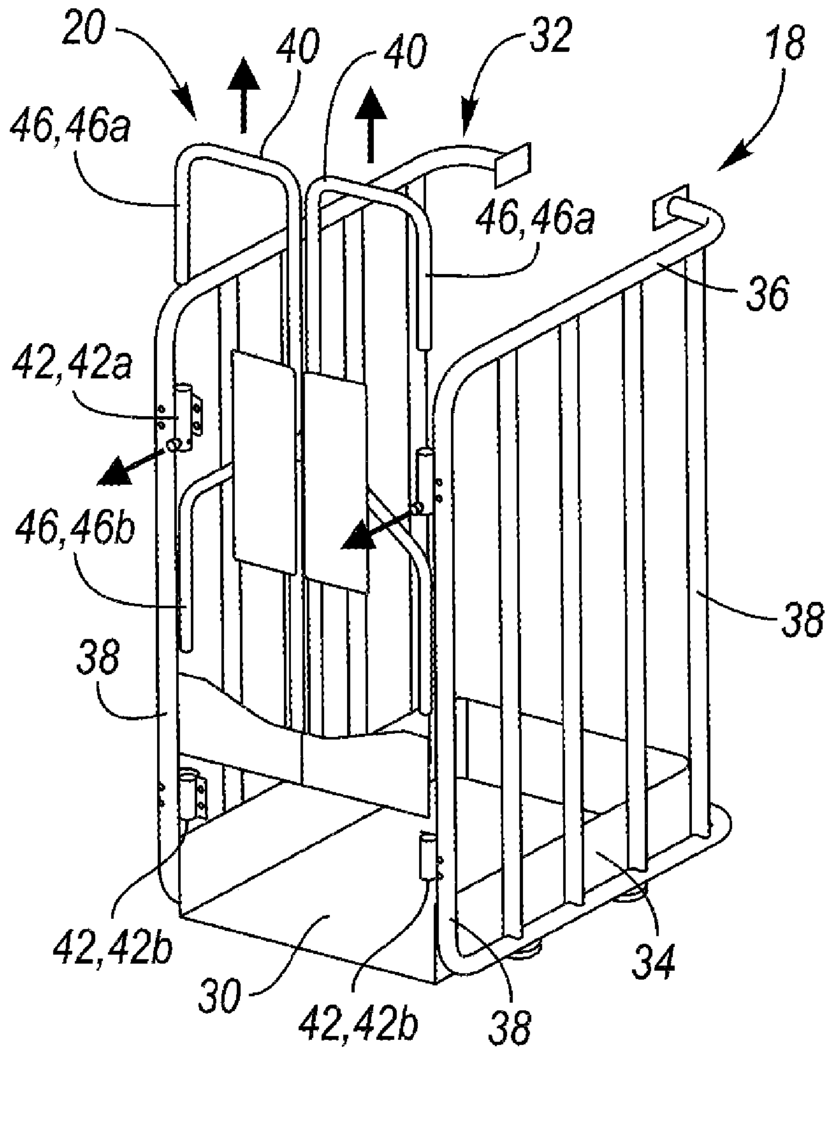
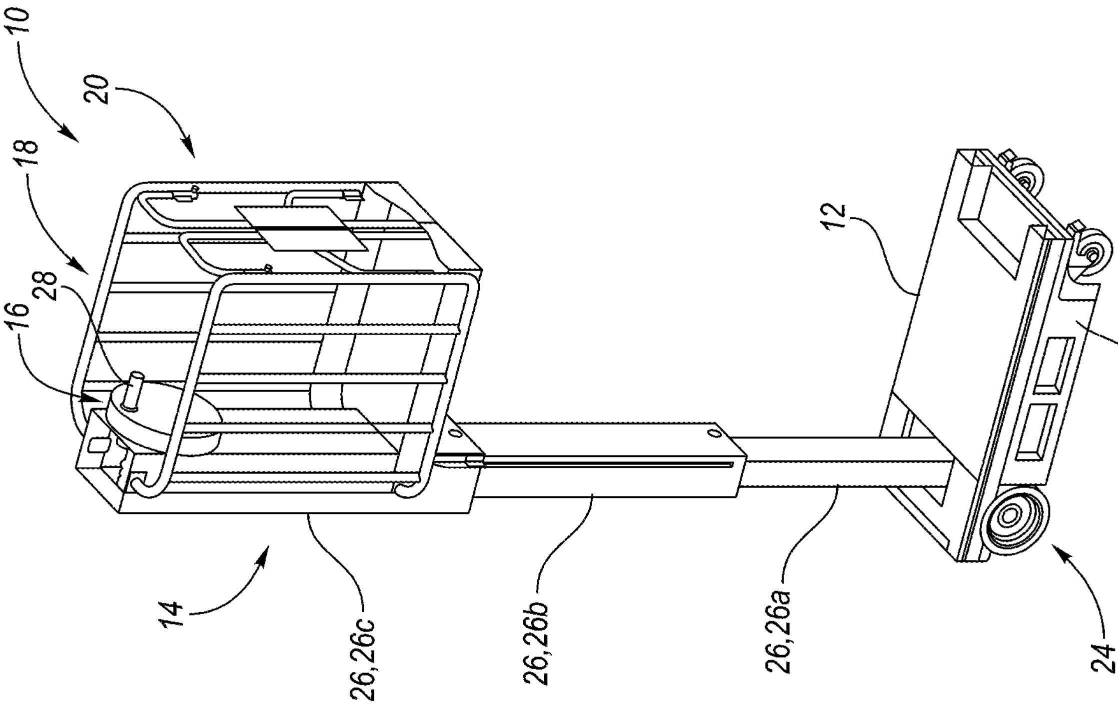
(57) **ABSTRACT**

A door assembly according to the disclosure may comprise a hinge; a door holder attached to the hinge, wherein the door holder defines a first opening; a door having a support that is receivable in the door holder, wherein the support defines a second opening that is alignable with the first opening when the support is positioned in the door holder; and a locking device that is insertable into the second opening when the support is positioned in the door holder and the second opening is aligned with the first opening to lock the door to the door holder. The locking device is removable from the second opening by pulling on at least a portion of the locking device so that the support may be removed from the door holder.




10
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18
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26,26c
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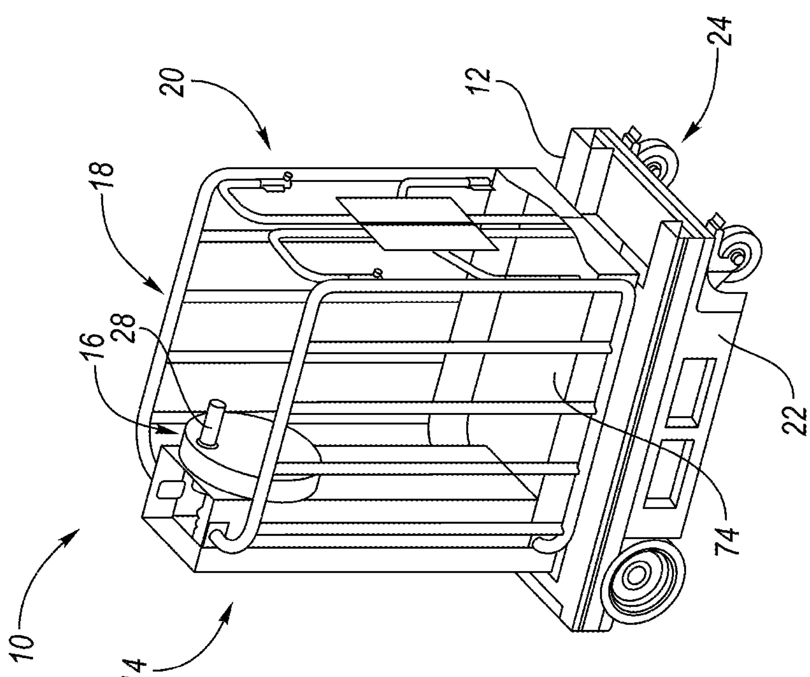
26,26a
24

FIG. 2


20
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74
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14

FIG. 1

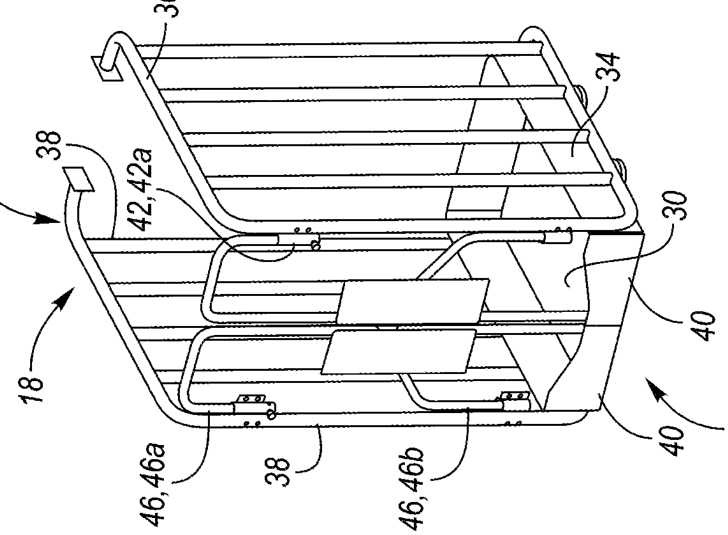

FIG. 3A

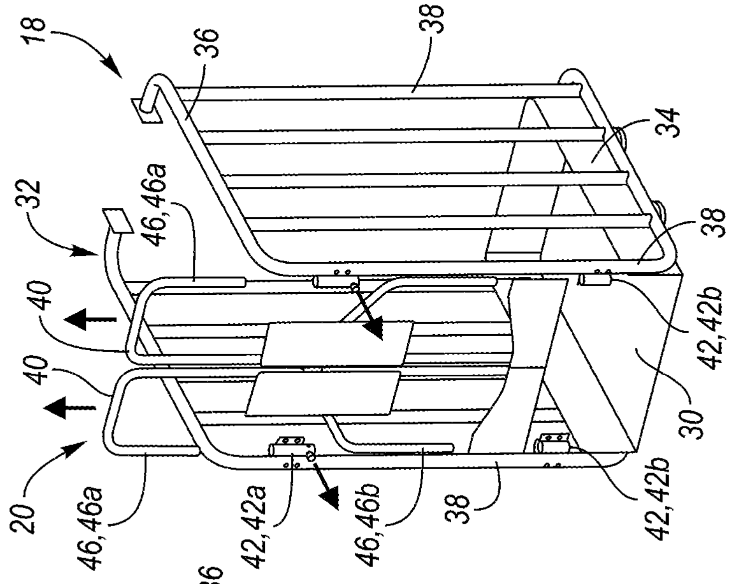

FIG. 3B

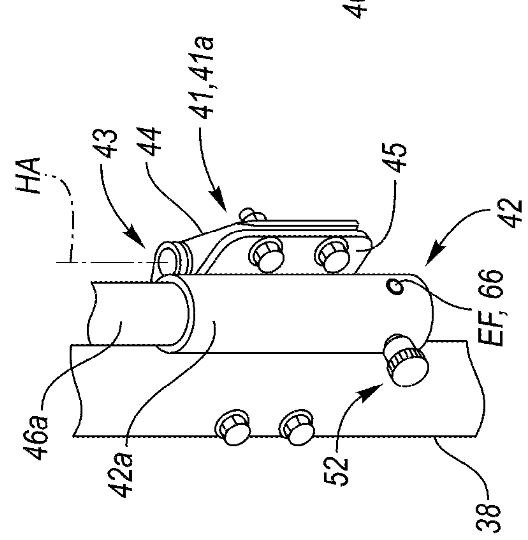

FIG. 4

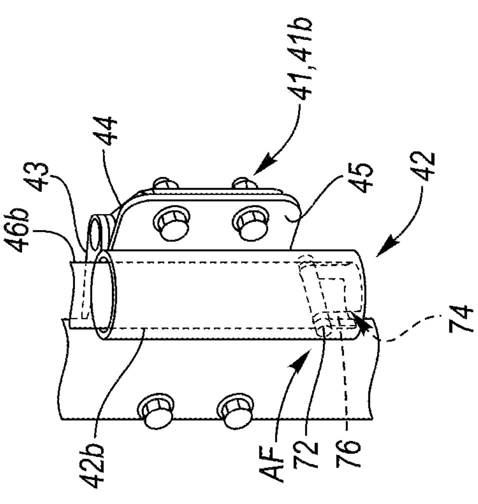

FIG. 5

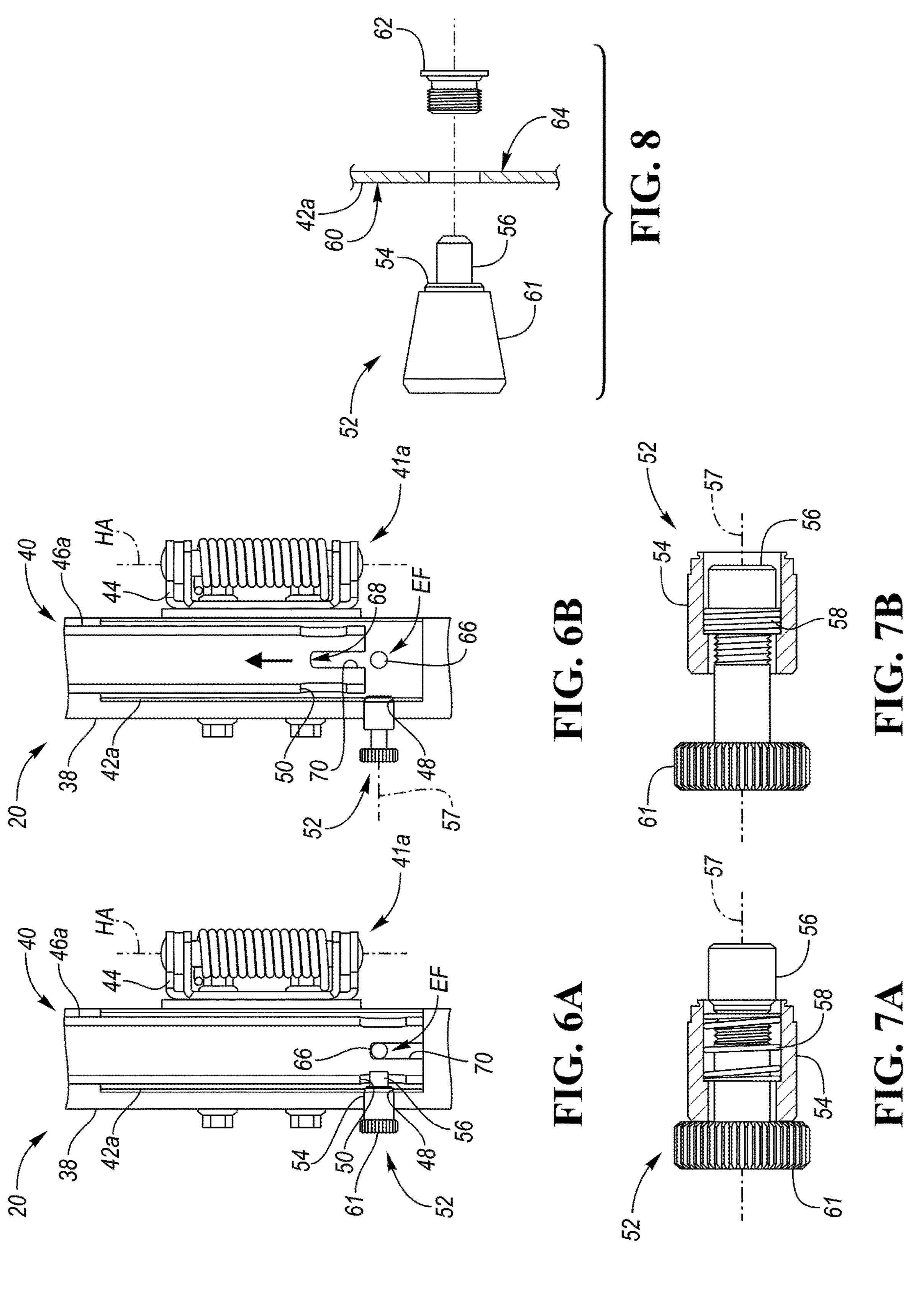

FIG. 6A

FIG. 6B

FIG. 7A

FIG. 7B

FIG. 8

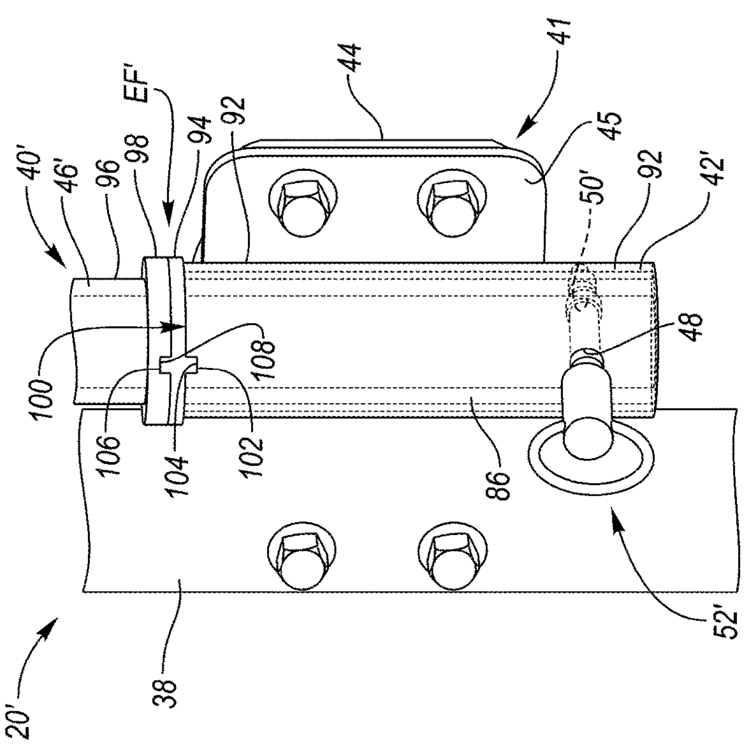

FIG. 9A

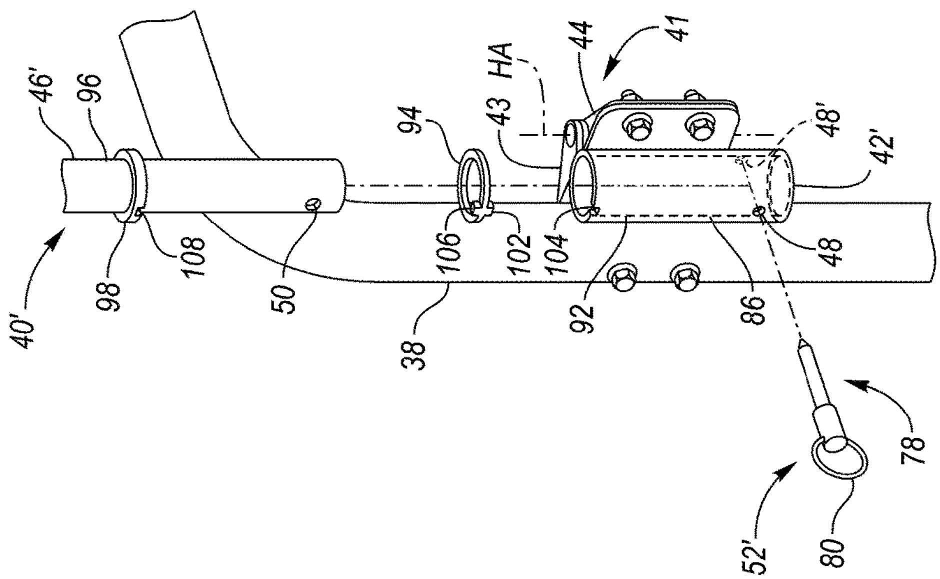

FIG. 9B

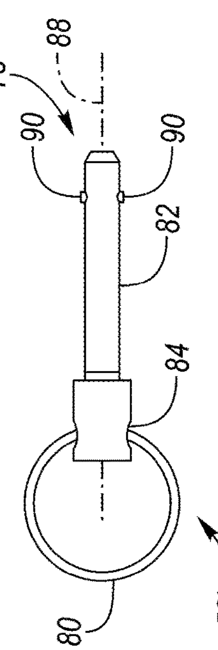

FIG. 10

DOOR ASSEMBLY

TECHNICAL FIELD

[0001] The disclosure relates to a door assembly.

BACKGROUND

[0002] An example door assembly is disclosed in U.S. Pat. No. 8,016,074 B2.

SUMMARY

[0003] A door assembly according to the disclosure may comprise a hinge; a door holder attached to the hinge, wherein the door holder defines a first opening; a door having a support that is receivable in the door holder, wherein the support defines a second opening that is alignable with the first opening when the support is positioned in the door holder; and a locking device that is insertable into the second opening when the support is positioned in the door holder and the second opening is aligned with the first opening to lock the door to the door holder. The locking device is removable from the second opening by pulling on at least a portion of the locking device so that the support may be removed from the door holder.

[0004] While exemplary embodiments are illustrated and disclosed, such disclosure should not be construed to limit the claims. It is anticipated that various modifications and alternative designs may be made without departing from the scope of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a perspective view of a lift with a work platform assembly including a door assembly according to disclosure, wherein the lift is shown in a lowered position;

[0006] FIG. **2** is a perspective view of the lift of FIG. **1** shown in a raised position;

[0007] FIG. **3**A is a perspective view of the work platform showing two doors of the door assembly each in a use position;

[0008] FIG. **3**B is a perspective view of the work platform of FIG. **3**A showing the doors unlocked from the work platform assembly and removed from the work platform assembly;

[0009] FIG. **4** is an enlarged perspective view of a first door holder of the door assembly and a first support of a door received in the first door holder, wherein the first door holder is attached to a first hinge which is attached to a frame member of a frame assembly of the work platform assembly;

[0010] FIG. **5** is an enlarged perspective view of a second door holder of the door assembly and a second support of the door received in the second door holder, wherein the second door holder is attached to a second hinge which is attached to the frame member of the frame assembly;

[0011] FIG. **6**A is a cross-sectional view of the first door holder and the first support of the door received in the first door holder, and showing a locking device in a locking position for securing the first support to the first door holder;

[0012] FIG. **6**B is a cross-sectional view similar to FIG. **6**A showing the locking device in a released position, and the first support of the door being removed from the first door holder;

[0013] FIG. **7**A is an enlarged side view of the locking device, partially in section, in the locking position;

[0014] FIG. **7**B is an enlarged side view of the locking device, partially in section, in the released position;

[0015] FIG. **8** is an enlarged side view of another embodiment of a locking device for locking the first support of the door to the first door holder;

[0016] FIG. **9**A is a perspective similar to FIG. **4** showing a portion of another embodiment of door assembly according to the disclosure, including additional example embodiments of a door holder, a support portion of a door received in the door holder, and a locking device for securing the support portion to the door holder;

[0017] FIG. **9**B is an exploded perspective view of the portion of the door assembly of FIG. **9**A showing the support portion of the door removed from the door holder, and the locking device removed from the support portion and the door holder; and

[0018] FIG. **10** is a side view of the locking device shown in FIGS. **9**A and **9**B.

DETAILED DESCRIPTION

[0019] As required, detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are merely exemplary, and that various and alternative forms may be employed. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art.

[0020] As used in the description of the various described embodiments and the appended claims, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including." "comprises" and/or "comprising" specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0021] FIGS. **1** and **2** show an example lift device or lift **10** according to the disclosure, such as a boom lift or aerial work platform. The lift **10** includes a base **12**, a support member arrangement **14** attached to the base **12**, a drive system **16** that is operable to move the support member arrangement **14** between a lowered position, shown in FIG. **1**, and a raised position, shown in FIG. **2**, and a work platform assembly **18** connected to the support member arrangement **14** and configured to receive an operator, wherein the work platform assembly **18** includes a novel door assembly **20** (e.g., gate assembly) according to the disclosure as explained below in detail.

[0022] The base **12** of the lift **10** shown in FIGS. **1** and **2** is configured to support the support member arrangement **14**, along with the rest of the lift **10**, and includes a main body **22**, such as a rigid support structure, and one or more support members **24**, such as wheels, tracks, outriggers, etc., attached to the main body **22**. In the illustrated embodiment, the lift **10** includes multiple support members **24** formed as wheels and associated tires, so that the base **12** is movable. Furthermore, two or more of the wheels are pivotable in order to steer the base **12**.

[0023] The support member arrangement **14** may include one or more movable support members **26**, such as pivotable and/or linearly movable booms, masts, tubes, columns, arms, etc., attached to the base **12**. In the illustrated embodiment, the support member arrangement **14** includes first, second and third upright support members **26***a*, **26***b* and **26***c*, respectively, and the second and third upright support members **26***b* and **26***c*, respectively, are linearly movable with respect to the first upright support member **26***a* when the support member arrangement **14** moves between the lowered and raise positions. In another embodiment, the support member arrangement **14** may include two upright support members, wherein one of the upright support members may be movable with respect to the other upright support member.

[0024] The drive system **16** may have any suitable configuration that is operable to move one or more of the support members **26** in order to move the support member arrangement **14** between the lowered and raised positions. For example, the drive system **16** may include one or more drive members, such as cylinders (e.g., hydraulic and/or electric cylinders), flexible drive members (e.g., strands, belts, chains and/or cables), rigid drive members (e.g., rack and pinion systems, rotatable shafts and gears), etc., and one or more actuators, such as automatic actuators (e.g., electric motors or pumps) and/or manually operable (i.e., hand-operated) actuators. In an embodiment including an automatic actuator, the actuator may be operable to move one or more drive members formed as cylinders between retracted and extended positions, for example. In the illustrated embodiment, the drive system **16** includes a manually operated actuator **28**, which may include a rotatable member, such as a wheel, disc, hand crank, lever arm, etc., that is configured to move one or more drive members, such as through transmission system including a gear system (not shown). Examples of a lift including a manual operable actuator or an automatic actuator are disclosed in U.S. application Ser. No. 18/593,237 (corresponding to Attorney Docket TERS 0147 PUS) filed on Mar. 1, 2024, which is hereby incorporated by reference in its entirety.

[0025] Referring to FIGS. **1**-**3**B, the work platform assembly **18** is movable with the support member arrangement **14** when the support member arrangement **14** moves between the lowered and raised positions. The work platform assembly **18** includes a base or work platform **30** (e.g., flat work platform) sized to receive and support an operator thereon, and a frame assembly **32**, such as a railing assembly, provided on the platform **30**. In the illustrated embodiment, the frame assembly **32** extends around, or partially around, a perimeter of the platform **30**, and includes multiple frame members, such as toe boards **34**, horizontal rails **36**, and vertical or upright rails **38**, which are connected to the toe boards **34** and the horizontal rails **36**. As mentioned above, the work platform assembly **18** also includes the door assembly **20**, which is connected to the frame assembly **32**. Furthermore, at least a portion of the frame assembly **32** may be considered part of the door assembly **20**.

[0026] Alternatively, the door assembly **20** may be used with any suitable lift having any suitable configuration, such as a scissor lift, articulated boom lift, straight boom lift, etc. Furthermore, any lift with which the door assembly **20** may be used may be a manual lift (e.g., manually operated lift) or an automatic lift, such as a hydraulic lift or electric lift.

[0027] While the door assembly **20** may include one door (e.g., gate) or multiple doors (e.g., gates), in the illustrated embodiment, the door assembly **20** includes two doors **40** that are each connected to a respective frame member, such as an upright rail **38** positioned adjacent the door **40** and which may be referred to as a side rail. Referring to FIGS. **3**A-**5**, the door assembly **20** also includes one or more hinges **41** connected to each upright rail **38** positioned adjacent to each of the doors **40**, and a door holder **42** connected to each of the hinges **41**. For example, each hinge **41** includes a fixed hinge part **43** fixedly connected to a respective upright rail **38**, such as with one more suitable fasteners (e.g., screws, bots, nuts, etc.), and a movable hinge part **44** that is pivotally connected to the fixed hinge part **43** so that the movable hinge part **44** is pivotable about a hinge axis HA of the hinge **41**. Furthermore, each door holder **42** is fixedly connected to a respective movable hinge part **44** so that the door holder **42** is movable (e.g., pivotable) with the hinge part **44** about the hinge axis HA. For example, each door holder **42** may include a projecting portion **45** (e.g., tab) that extends from a main body of the door holder **42** and is connected to the hinge part **44** with one or more suitable fasteners (e.g., screws, bots, nuts, etc.).

[0028] In the illustrated embodiment, the door assembly **20** includes two hinges **41** and two door holders **42** for each door **40**, and each door **40** is movable (e.g., pivotable) between a closed position shown in FIG. **3**A and an open position (not shown) for allowing access to the work platform assembly **18**. More specifically, the door assembly **20** includes a first or upper hinge **41***a* and an associated first or upper door holder **42***a*, and a second or lower hinge **41***b* and an associated second or lower door holder **42***b* for each door **40**, and each door **40** is pivotable inwardly from the closed position to the open position. Furthermore, each door **40** includes multiple support portions or supports **46** that are each receivable in one of the door holders **42**. For example, each door **40** may include an upper support **46***a* that is insertable into or receivable in an upper door holder **42***a*, and a lower support **46***b* that is insertable into or receivable in a lower door holder **42***b*.

[0029] One or both of the door holders **42** for each door **40** may be configured as a quick-release door holder. In the illustrated embodiment, for example, each upper door holder **42***a* is configured as a quick-release door holder. In that regard, referring to FIGS. **6**A-**6**B, each upper door holder **42***a* defines a first opening **48** that is alignable with a second opening **50** formed in a respective upper support **46***a* of a respective door **40**, and the door assembly **20** further includes a locking device **52** that is insertable into the second opening **50** (e.g., movable to a locking position shown in FIG. **6**A) when the upper support **46***a* is positioned in the upper door holder **42***a* and the second opening **50** is aligned with the first opening **48** to lock the upper support **46***a* and the associated door **40** to the upper door holder **42***a*. Furthermore, referring to FIGS. **3**B and **6**B, the locking device **52** is removable from the second opening **50** (e.g., movable to a released or retracted position) by pulling on the locking device **52**, or a portion of the locking device **52**, so that the upper support **46***a* may be removed from the upper door holder **42***a* to thereby remove the door **40** from the work platform assembly **18**.

[0030] In the embodiment shown in FIGS. **6**A-**7**B, each locking device **52** comprises a first portion, such as an outer body **54**, that is engageable with the upper door holder **42***a*, a second portion, such as an inner portion **56** (e.g., a rod, pin, plunger, etc.) configured for insertion into the second opening **50**, and a biasing member **58**, such as a coil spring, helical spring, or any other suitable spring, that urges the inner portion **56** into the second opening **50** when the upper support **46***a* is positioned in the upper door holder **42***a*. The inner portion **56** is movable along a longitudinal axis **57** of the locking device **52** between a locking position in which the inner portion **56** is inserted into the second opening **50** as shown in FIG. **6**A, and a released or retracted position in which the inner portion **56** is removed from the second opening **50** as shown in FIG. **6**B. Each locking device **52** may also include a grip portion **61** attached to the inner portion **56**, and the grip portion **61** may be gripped by a user and pulled away from the upper door holder **42***a* in order to move the inner portion **56** from the locking position to the released position. For example, each locking device **52** may be formed as a spring-loaded plunger.

[0031] Furthermore, the outer body **54** may be secured to the upper door holder **42***a* in any suitable manner. In the embodiment shown in FIG. **6**A-**7**B, for example, the outer body **54** is configured to be press fit into the first opening **48** of the door holder **42***a* so that the outer body **54** is connected to the upper door holder **42***a* by an interference fit. Furthermore, the outer body **54** may be made of any suitable material, such as metal (e.g., steel, stainless steel, aluminum, etc.), plastic or a composite material, configured to form an interference fit (e.g., configured to plastically deform) when the outer body **54** is pressed into the first opening **48** of the door holder **42***a*. As another example, the outer body **54** may be fastened to the upper door holder **42***a* with a fastener. In the embodiment shown in FIG. **8**, for example, the outer body **54** is engageable with an outer surface **60** of the upper door holder **42***a*, the locking device **52** further includes a sleeve **62** that is engageable with an inner surface **64** of the upper door holder **42***a*, and the sleeve **62** may be threadingly connectable to the outer body **54** to secure the outer body **54** to the upper door holder **42***a*.

[0032] Returning to FIGS. **6**A and **6**B, the door assembly **20** may be configured so that, for each door **40**, the upper door holder **42***a* and the upper support **46***a* of the door **40** each comprise an engagement feature EF, and the engagement features EF are engageable with each other when the upper support **46***a* is positioned in the upper door holder **42***a* so that the upper door holder **42***a* supports at least a portion of a weight or mass of the door **40**. With such a configuration, the door assembly **20** may be configured so that, for each door **40**, no weight or mass of the door **40** rests on the associated locking device **52** when the locking device **52** is inserted into the second opening **50** of the upper support **46***a*. As a result, the locking device **52** may be easily removed from the second opening **50** by pulling outwardly on at least a portion (e.g., the grip portion **61**) of the locking device **52**. For example, the engagement feature EF of one of the upper door holder **42***a* and the upper support **46***a* may comprise an engagement member, such as a projection, pin, etc., and the engagement feature EF of the other of the upper door holder **42***a* and the upper support **46***a* may comprise a receptacle, such as a slot, notch, groove, etc., that receives the engagement member when the upper support **46***a* is positioned in the upper door holder **42***a*. In the embodiment shown in FIGS. **6**A and **6**B, the engagement feature EF of the upper door holder **42***a* comprises a pin **66** positioned proximate a lower end of the upper door holder **42***a*, and the engagement feature EF of the upper support **46***a* comprises an engagement surface **68** that at least partially defines a slot **70** at a lower end of the upper support **46***a*, wherein the slot **70** is configured to receive the pin **66** and the engagement surface **68** is engageable with the pin **66** when the upper support **46***a* is positioned in the upper door holder **42***a*. In the embodiment shown in FIGS. **4**, **6**A and **6**B, the pin **66** is oriented in a direction transverse to the longitudinal axis **57** of the locking device **52** when the upper support **46***a* is positioned in the upper door holder **42***a* and the locking device **52** is inserted the second opening **50**.

[0033] For each door **40**, the engagement features EF of the upper door holder **42***a* and the upper support **46***a* may also, or instead, function as alignment features. In that regard, the engagement features EF may be cooperable with each other to align the first opening **48** and the second opening **50** when the upper support **46***a* is positioned in the upper door holder **42***a*. In addition, the engagement features EF may be cooperable with each other so that the upper door holder **42***a* and the upper support **46***a* are movable together with respect to the hinge axis HA of the associated hinge **41***a*. In the embodiment shown in FIGS. **6**A and **6**B, for example, the pin **66** and the engagement surface **68**, which defines the slot **70**, cooperate with each other to align the openings **48** and **50** when the upper support **46***a* is positioned in the upper door holder **42***a*. Likewise, the pin **66** and the engagement surface **68** are engageable with each other so as to fix together the upper door holder **42***a* and the upper support **46***a* so that the upper door holder **42***a* and the upper support **46***a* are movable, e.g., pivotable, together along with the hinge part **44** and with respect to the hinge axis HA of the associated hinge **41***a*.

[0034] Referring to FIGS. **3**A, **3**B and **5**, the door assembly **20** may be configured so that, for each door **40**, the lower door holder **42***b* and the lower support **46***b* of the door **40** each comprise an alignment or securing feature AF, and the alignment features AF are cooperable with each other when the lower support **46***b* is positioned in the lower door holder **42***b* so that the lower door holder **42***b* and the lower support **46***b* are movable (e.g., pivotable) together along with the associated hinge part **44** and with respect to the hinge axis HA of the associated hinge **41***b*. For example, the alignment feature AF of one of the lower door holder **42***b* and the lower support **46***b* may comprise an alignment member, such as a projection, pin, etc., and the alignment feature AF of the other of the lower door holder **42***b* and the lower support **46***b* may comprise a receptacle, such as a slot, notch, groove, etc., that receives the alignment member when the lower support **46***b* is positioned in the lower door holder **42***b*. In the embodiment shown in FIG. **5**, the alignment feature AF of the lower door holder **42***b* comprises a pin **72** positioned proximate a lower end of the lower door holder **42***b*, and the alignment feature AF of the lower support **46***b* comprises an engagement surface **74** that at least partially defines a slot **76** at a lower end of the lower support **46***b*, wherein the slot **76** is configured to receive the pin **72** and the engagement surface **74** is engageable with the pin **72** when the lower support **46***b* is positioned in the lower door holder **42***b*.

[0035] For each door **40**, the alignment features AF of the lower door holder **42***b* and the lower support **46***b* may also be configured to support at least a portion of the weight or mass of the door **40**.

[0036] With the above configuration, at least one of the door holders **42** for each door **40** of the door holder assembly

20 may be configured to cooperate with the respective support **46** of the door **40** to support at least a portion of the weight or mass of the door **40** so that no weight or mass of the door **40** is supported by the locking device **52**. For example, the door holder **42***a* or the door holder **42***b* may support all of the weight or mass of the door **40**, or each door holder **42***a*, **42***b* may support a portion of the weight or mass of the door **40** so that the door holders **42***a* and **42***b* together support all of the weight or mass of the door **40**.

[0037] Furthermore, in another embodiment, the door assembly **20** may be configured so that the locking device **52** is provided with the lower door holder **42***b*. In yet another embodiment, a locking device **52** may be provided for each door holder **42***a* and **42***b*. In still yet another embodiment, the door assembly **20** may include a single door holder **42** for each door **40**, and a single locking device **52** for use with the single door holder **42** to lock a support portion of the door **40** to the single door holder **42**.

[0038] FIGS. **9**A and **9**B show a portion of another embodiment **20'** of a door assembly (e.g., gate assembly) according to the disclosure. The door assembly **20'** is similar to the door assembly **20** and may include the same or similar features as the door assembly **20**, which features are identified with the same or similar reference numerals, but the reference numerals used to identify similar features in the door assembly **20'** may each include a prime mark. In that regard, the door assembly **20'** may include one or more doors **40'** (e.g., gates) that are each connected to a respective frame member, such as an upright rail **38** positioned adjacent the door **40'**. Furthermore, each door **40'** may be connected to the respective upright rail **38** with one or more door holders **42'**. The portion of the door assembly **20'** shown in FIGS. **9**A and **9**B includes a hinge **41** connected to the upright rail **38**, an example door holder **42'**, which may be an upper door holder or a lower door holder if the door assembly **20'** includes two door holders **42'** per door **40'**, and a support portion or support **46'** of a door **40'** connected to the door holder **42'**. In addition, the portion of the door assembly **20'** shown in FIGS. **9**A and **9**B includes another embodiment **52'** of a locking device that is insertable into a first opening **48** of the door holder **42'** and a second opening **50** of the support **46'** (e.g., movable to or positionable in a locking position shown in FIG. **9**A), when the openings **48** and **50** are aligned with each other, in order to lock the support **46'** to the door holder **42'**. The locking device **52'** may further be extendable through an additional first opening **48'** of the door holder **42'**, which is positioned opposite the first opening **48**, and an additional second opening **50'** of the support **46'**, which is positioned opposite the second opening **50**. With such a configuration, the locking device **52'** may extend through the door holder **42'** and the support **46'** when the locking device **52'** is in the locking position.

[0039] Referring to FIGS. **9**A-**10**, the locking device **52'** includes a locking pin **78** and a pull member **80**, such as a pull tab or ring, for removing the locking device **52'** (e.g., the locking pin **78**) from the openings **48**, **48'**, **50**, **50'** so that the support **46'** may be removed from the door holder **42'**. The illustrated locking pin **78** comprises a pin body **82** having an enlarged head **84** that may be engageable with an outer surface **86** of the door holder **42'** when the locking device **52'** is in the locking position, and the pin body **82** may have a longitudinally extending pin axis **88**. Furthermore, the illustrated locking pin **78** includes one or more lock elements **90**, such as projections, balls, beads, etc., that are each movably associated with the pin body **82** so that the lock elements **90** are each movable in a direction transverse to the pin axis **88**. In addition, the locking pin **78** includes one or more biasing members, such as coil or helical springs (not shown), positioned in the body **82** and configured to urge the lock elements **90** to each move in the direction transverse to the pin axis **88**. With such a configuration, each lock element **90** may be biased outwardly by the one or more biasing members and away from the pin axis **88** when the locking device **52'** is inserted in the openings **48**, **48'**, **50**, **50'** to hold the locking device **52'** in the locking position, and each lock element **90** may move inwardly toward the pin axis **88**, and against the bias of the one or more biasing members, when the locking device **52'** is removed from the openings **48**, **48'**, **50**, **50'**.

[0040] In the embodiment shown in FIGS. **9**A and **9**B, the door holder **42'** and the support **46'** of the door **40'** each comprise an engagement feature EF', and the engagement features EF' are engageable with each other when the support **46'** is positioned in the door holder **42'** so that the door holder **42'** supports at least a portion of a weight or mass of the door **40'** so that the locking device **52'** does not support any weight or mass of the door **40'**. With such a configuration, the locking device **52'** may be easily inserted into the openings **48**, **48'**, **50**, **50'** to secure the door **40'** to the door holder **42'**, and the locking device **52'** may be easily removed from the openings **48**, **48'**, **50**, **50'** when it is desired to quickly remove the door **40'** from the door holder **42'**, for example. In the illustrated embodiment, the door holder **42'** comprises a door holder body **92**, and the engagement feature EF' of the door holder **42'** comprises an engagement member **94**, such as a washer, thrust washer, plate, etc., associated with the door holder body **92** (e.g., positionable on the door holder body **92** and/or attached to the door holder body **92**). Likewise, the support **46'** comprises a support body **96** and the engagement feature EF' of the support **46'** comprises a support member **98**, such as a projection (e.g., tab, plate, etc.), that projects outwardly (e.g., radially outwardly) from the support body **96**. For example, the support member **98** may be a plate, such as a circular or round plate, welded or otherwise attached to the support body **96**. As another example, the engagement feature EF' of the door holder **42'** may be a surface **100** (e.g., top surface) of the door holder body **92** that is directly engageable with the support member **98** of the support **46'**.

[0041] The engagement features EF' of the door holder **42'** and the support **46'** may also each include one or more alignment features for aligning the support **46'** with respect to the door holder **42'** (e.g., for aligning the openings **48**, **48'**, **50**, **50'** with respect to each other) and/or for fixing the support **46'** with respect to the door holder **42'** so that the support **46'** and the door holder **42'** are pivotable together and with the movable hinge part **44** about the hinge axis HA of the hinge **41**. For example, the door holder body **92** and the engagement member **94** of the door holder **42'** may each include a first alignment feature, and the first alignment features may be cooperable with each other for fixing the engagement member **94** with respect to the door holder body **92**. In addition, the engagement member **94** of the door holder **42'** and the support member **98** of the support **46'** may each include a second alignment feature, and the second alignment features may be cooperable with each other for aligning the support **46'** with respect to the door holder **42'** and for fixing position of the support **46'** with respect to the door holder **42'** so that the support **46'** and the door holder **42'** are movable together. As a more detailed example, the door holder body **92** may include one of a first projection **102** (e.g., ridge, tab, pin, etc.) and a first receptacle **104** (e.g., recess, notch, groove, aperture, etc.), and the engagement member **94** may include the other of the first projection **102** and the first receptacle **104**, and the first projection **102** and the first receptacle **104** may be engageable or otherwise cooperable with each other for fixing the engagement member **94** with respect to the door holder body **92**. Likewise, the engagement member **94** may include one of a second projection **106** (e.g., ridge, tab, pin, etc.) and a second receptacle **108** (e.g., recess, notch, groove, aperture, etc.), and the support member **98** may include the other of the second projection **106** and the second receptacle **108**, and the second projection **106** and the second receptacle **108** may be engageable or otherwise comparable with each other for aligning the support **46'** with respect to the door holder **42'** and for fixing position of the support **46'** with respect to the door holder **42'** so that the support **46'** and the door holder **42'** are movable together.

[0042] As another example, the locking device **52'** may be configured to fix position of the support **46'** with respect to the door holder **42'** so that the support **46'** and the door holder **42'** are movable together. In that regard, when the locking pin **78** is inserted in the openings **48**, **48'**, **50**, **50'**, the locking pin **78** may engage the door holder **42'** and the support **46'** and fix position of the support **46'** with respect to the door holder **42'**.

[0043] As yet another example, the door **40'** may include an additional support and the door assembly **20'** may include an additional door holder configured to receive the additional support so that the additional door holder supports at least a portion of the weight or mass of the door **40'**. In that regard, the additional door holder and the additional support may include any of the engagement features EF, EF' discussed above in detail. Therefore, like the door assembly **20** described above, at least one of the door holder **42'** and the additional door holder of the door assembly **20'** may be configured to cooperate with the support **46'** or the additional support of the door **40'** to support at least a portion of the weight or mass of the door **40'** so that no weight or mass of the door **40'** is supported by the locking device **52'**. For example, the door holder **42'** or the additional door holder may support all of the weight or mass of the door **40'**, or each of the door holder **42'** and the additional door holder may support a portion of the weight or mass of the door **40'**. Furthermore, the additional support and the additional door holder may include any of the alignment features described above for fixing position of the additional support with respect to the additional door holder.

[0044] With any of the above described door assembly embodiments, one or more doors may each be quickly and easily removed from an associated door holder by removing an associated locking device from an opening formed in a portion of the door (e.g., by simply pulling on at least a portion of the locking device) so that the door may be removed from the door holder. Likewise, each door may be quickly and easily reinstalled in the associated door holder by positioning the portion of the door in the associated door holder and reinserting the associated locking device into the opening of the door so that the door may be locked in a use position with respect to the door holder.

[0045] The following clauses describe aspects of embodiments according to the disclosure.

[0046] Clause 1. A door assembly according to the disclosure may comprise a hinge; a door holder attached to the hinge, wherein the door holder defines a first opening; a door having a support that is receivable in the door holder, wherein the support defines a second opening that is alignable with the first opening when the support is positioned in the door holder; and a locking device that is insertable into the second opening when the support is positioned in the door holder and the second opening is aligned with the first opening to lock the door to the door holder. The locking device is removable from the second opening by pulling on at least a portion of the locking device so that the support may be removed from the door holder.

[0047] Clause 2. The door assembly of the preceding clause wherein the locking device comprises a first portion that is engageable with the door holder, a second portion for insertion into the second opening, and a biasing member that urges the second portion into the second opening when the support is positioned in the door holder.

[0048] Clause 3. The door assembly of clause 2 wherein the first portion of the locking device is configured to be press fit into the first opening of the door holder.

[0049] Clause 4. The door assembly of any of the preceding clauses wherein the locking device comprises a locking pin that is insertable into the first and second openings when the support is positioned in the door holder and the second opening is aligned with the first opening.

[0050] Clause 5. The door assembly of clause 4 wherein the locking pin comprises a pin body having a pin axis, a lock element that is movably associated with the pin body so that the lock element is movable in a direction transverse to the pin axis.

[0051] Clause 6. The door assembly of any of the preceding clauses further comprising a frame member attached to the hinge, wherein the door has a mass, the door holder and the support of the door each comprise an engagement feature, and the engagement features are engageable with each other when the support is positioned in the door holder so that the door holder supports at least a portion of the mass of the door.

[0052] Clause 7. The door assembly of clause 6 wherein the engagement feature of one of the door holder and the support comprises an engagement member, and the engagement feature of the other of the door holder and the support comprises a receptacle that receives the engagement member when the support is positioned in the door holder.

[0053] Clause 8. The door assembly of clause 6 wherein the engagement feature of the door holder comprises a pin positioned proximate a lower end of the door holder, and the engagement feature of the support comprises an engagement surface that at least partially defines a slot at a lower end of the support, and wherein the slot is configured to receive the pin and the engagement surface is engageable with the pin when the support is positioned in the door holder.

[0054] Clause 9. The door assembly of any of clauses 6-8 where the engagement features are further cooperable with each other to align the first opening and the second opening when the support is positioned in the door holder.

[0055] Clause 10. The door assembly of clause 6 wherein the engagement feature of the door holder comprises a surface of the door holder, the support comprises a support body and the engagement feature of the support comprises a support member that projects outwardly from the support body.

[0056] Clause 11. The door assembly of clause 6 wherein the door holder comprises a door holder body, the engagement feature of the door holder comprises an engagement member associated with the door holder body, the support comprises a support body and the engagement feature of the support comprises a support member attached to the support body.

[0057] Clause 12. The door assembly of clause 11 wherein the engagement member of the door holder comprises a washer, and the support member of the support comprises a plate welded to the support body.

[0058] Clause 13. The door assembly of any of clauses 6-12 further comprising an additional hinge attached to the frame member and an additional door holder attached to the additional hinge, and the door comprises an additional support that is receivable in the additional door holder.

[0059] Clause 14. The door assembly of clause 13 wherein the additional door holder and the additional support each comprise an alignment feature, and the alignment features are cooperable with each other when the additional support is positioned in the additional door holder so that the additional door holder and the additional support are movable together with respect to a hinge axis of the additional hinge.

[0060] Clause 15. The door assembly of any of clauses 13 and 14 wherein the additional hinge is positioned on the frame member below the hinge, and the additional door holder is positioned below the door holder.

[0061] Clause 16. The door assembly of any of the preceding clauses wherein the door holder and the support of the door each comprise an alignment feature, and the alignment features are cooperable with each other when the support is positioned in the door holder to align the first and second openings and so that door holder and the support are movable together with respect to a hinge axis of the hinge.

[0062] Clause 17. The door assembly of clauses 16 wherein the alignment feature of one of the door holder and the support comprises an alignment member, and the alignment feature of the other of the door holder and the support comprises a receptacle that is configured to receive the alignment member when the support is positioned in the door holder.

[0063] Clause 18. The door assembly of clause 17 wherein the alignment member is oriented in a direction transverse to a longitudinal axis of the locking device when the support is positioned in the door holder and the locking device is inserted in the second opening.

[0064] Clause 19. A lift comprising a work platform including a frame assembly having a frame member; and the door assembly of any of the preceding clauses, wherein the hinge of the door assembly is attached to the frame member.

[0065] Clause 20. The lift of claim **19** further comprising an additional hinge attached to the frame member and an additional door holder attached to the additional hinge, and the door comprises an additional support that is receivable in the additional door holder, wherein the door has a mass, the door holder and the support each comprise an engagement feature, and the engagement features are engageable with each other when the support is positioned in the door holder so that the door holder at least partially supports the mass of the door and so that the locking device does not support any of the mass of the door when the locking device is inserted in the second opening of the support. Furthermore, the additional door holder and the additional support each comprise an alignment feature, and the alignment features are cooperable with each other when the additional support is positioned in the additional door holder so that the additional door holder and the additional support are movable together with respect to a hinge axis of the additional hinge.

[0066] Clause 21. Any of the preceding clauses 1-20 in any combination.

[0067] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms according to the disclosure. In that regard, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure. Additionally, the features of various implementing embodiments may be combined to form further embodiments according to the disclosure. For example, any features of the door assembly **20** may be used with the door assembly **20'**, and any features of the door assembly **20'** may be used with the door assembly **20**.

What is claimed is:

1. A door assembly comprising:

a hinge;

a door holder attached to the hinge, wherein the door holder defines a first opening;

a door having a support that is receivable in the door holder, wherein the support defines a second opening that is alignable with the first opening when the support is positioned in the door holder; and

a locking device that is insertable into the second opening when the support is positioned in the door holder and the second opening is aligned with the first opening to lock the door to the door holder, wherein the locking device is removable from the second opening by pulling on at least a portion of the locking device so that the support may be removed from the door holder.

2. The door assembly of claim **1** wherein the locking device comprises a first portion that is engageable with the door holder, a second portion for insertion into the second opening, and a biasing member that urges the second portion into the second opening when the support is positioned in the door holder.

3. The door assembly of claim **2** wherein the first portion of the locking device is configured to be press fit into the first opening of the door holder.

4. The door assembly of claim **1** wherein the locking device comprises a locking pin that is insertable into the first and second openings when the support is positioned in the door holder and the second opening is aligned with the first opening.

5. The door assembly of claim **4** wherein the locking pin comprises a pin body having a pin axis, a lock element that is movably associated with the pin body so that the lock element is movable in a direction transverse to the pin axis.

6. The door assembly of claim **1** further comprising a frame member attached to the hinge, wherein the door has a mass, the door holder and the support of the door each comprise an engagement feature, and the engagement features are engageable with each other when the support is positioned in the door holder so that the door holder supports at least a portion of the mass of the door.

7. The door assembly of claim **6** wherein the engagement feature of one of the door holder and the support comprises an engagement member, and the engagement feature of the other of the door holder and the support comprises a receptacle that receives the engagement member when the support is positioned in the door holder.

8. The door assembly of claim **6** wherein the engagement feature of the door holder comprises a pin positioned proximate a lower end of the door holder, and the engagement feature of the support comprises an engagement surface that at least partially defines a slot at a lower end of the support, and wherein the slot is configured to receive the pin and the engagement surface is engageable with the pin when the support is positioned in the door holder.

9. The door assembly of claim **6** where the engagement features are further cooperable with each other to align the first opening and the second opening when the support is positioned in the door holder.

10. The door assembly of claim **6** wherein the engagement feature of the door holder comprises a surface of the door holder, the support comprises a support body and the engagement feature of the support comprises a support member that projects outwardly from the support body.

11. The door assembly of claim **6** wherein the door holder comprises a door holder body, the engagement feature of the door holder comprises an engagement member associated with the door holder body, the support comprises a support body and the engagement feature of the support comprises a support member attached to the support body.

12. The door assembly of claim **11** wherein the engagement member of the door holder comprises a washer, and the support member of the support comprises a plate welded to the support body.

13. The door assembly of claim **6** further comprising an additional hinge attached to the frame member and an additional door holder attached to the additional hinge, and the door comprises an additional support that is receivable in the additional door holder.

14. The door assembly of claim **13** wherein the additional door holder and the additional support each comprise an alignment feature, and the alignment features are cooperable with each other when the additional support is positioned in the additional door holder so that the additional door holder and the additional support are movable together with respect to a hinge axis of the additional hinge.

15. The door assembly of claim **13** wherein the additional hinge is positioned on the frame member below the hinge, and the additional door holder is positioned below the door holder.

16. The door assembly of claim **1** wherein the door holder and the support of the door each comprise an alignment feature, and the alignment features are cooperable with each other when the support is positioned in the door holder to align the first and second openings and so that door holder and the support are movable together with respect to a hinge axis of the hinge.

17. The door assembly of claim **16** wherein the alignment feature of one of the door holder and the support comprises an alignment member, and the alignment feature of the other of the door holder and the support comprises a receptacle that is configured to receive the alignment member when the support is positioned in the door holder.

18. The door assembly of claim **17** wherein the alignment member is oriented in a direction transverse to a longitudinal axis of the locking device when the support is positioned in the door holder and the locking device is inserted in the second opening.

19. A lift comprising:

a work platform including a frame assembly having a frame member; and

the door assembly of claim **1**, wherein the hinge of the door assembly is attached to the frame member.

20. The lift of claim **19** further comprising an additional hinge attached to the frame member and an additional door holder attached to the additional hinge, and the door comprises an additional support that is receivable in the additional door holder, wherein the door has a mass, the door holder and the support each comprise an engagement feature, and the engagement features are engageable with each other when the support is positioned in the door holder so that the door holder at least partially supports the mass of the door and so that the locking device does not support any of the mass of the door when the locking device is inserted in the second opening of the support, and wherein the additional door holder and the additional support each comprise an alignment feature, and the alignment features are cooperable with each other when the additional support is positioned in the additional door holder so that the additional door holder and the additional support are movable together with respect to a hinge axis of the additional hinge.

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